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When workers get to choose: Employment and wage responses to demand shocks in labour-managed and conventional firms

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Abstract

In conventional firms (CFs), workers are unlikely to accept pay and hour reductions in order to secure their jobs, in particular because of information asymmetry. A specific type of firm is not subject to this information asymmetry problem because workers make decisions and share profits: worker cooperatives. In this paper, we use an exhaustive employer–employee panel French dataset in order to compare systematically adjustment to demand shocks in worker cooperatives and CFs. We find that, when faced with the same variation of demand, worker cooperatives have more stable employment and more variable pay. However, we show that this employment stability in cooperatives benefits primarily men and employees in low-skill jobs.

KEYWORDS
cooperative, demand shock, employment, hours, wages

JEL CLASSIFICATION
J54, D21, P13, J23, J63

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1 | INTRODUCTION

All firms face a flexibility challenge—in order to survive economic fluctuations, they must be able to adjust their production capacities and their costs. Overall, firms' primary response to demand shocks seems to be to adjust employment, as was observed, for example, in many countries during the recession that followed the 2008 financial crisis (Bertola et al., 2012; Fabiani et al., 2015; Izquierdo et al., 2017; Tijdens et al., 2014). More efficient solutions might take into account employee preferences and involve other responses such as pay adjustments or hour adjustments with some employment protection in downturns (Bertheau et al., 2022; Davis & Krolikowski, 2025). The theoretical literature identifies one key barrier: information asymmetry between employers and employees (Aoki, 1984) which prevents workers from accepting temporary wage reductions in order to secure their employment, because they may not know whether such cuts are warranted and employers have incentives to renege on such agreements. In this paper, we examine what happens when workers get to decide, as is the case in worker cooperatives, and whether adjustments in response to demand shocks are different.

A few important studies after the Great Recession showed some diversity in firms' responses, which appear to be determined by the institutional context (Bassanini & Garnero, 2013; Ellul et al., 2018; Fabiani et al., 2015), ownership structure (Bassanini et al., 2013; D'Aurizio & Romano, 2013; Ellul et al., 2018), including the presence of broad-based Employee Stock Ownership Plans (ESOPs—see Kurtulus & Kruse, 2018) and employee representation (Gregorič & Rapp, 2019). Recent research has also shown that job cuts do not affect all groups of employees equally, even accounting for industry and occupational segregation (Alon et al., 2020; Périvier, 2014; Razzu & Singleton, 2018). Many articles study firms' responses to economic fluctuations more generally (Di Maggio et al., 2020; Sockin & Sockin, 2025). However, microeconomic evidence about the effect of firms' governance on their adjustment strategies is still limited, particularly when it comes to the influence of employees' participation in governance.

Employee participation in governance is thought to allow conventional firms (CFs) to overcome informational barriers that prevent efficient solutions to be selected by creating trust and cooperation (see, e.g., Rogers & Streek, 1995). At the board level, it has been shown to protect employment and increase pay flexibility in Scandinavian companies (Gregorič & Rapp, 2019) and in German firms with parity co-determination (Kim et al., 2018). In labour-managed firms (LMFs), employee owners approve all major decisions and elect senior management and directors. As a result, information asymmetry is greatly decreased, employee preferences are known, and any pay cut associated with employment preservation is fully incentive compatible—as employees make the decisions and claim future profits. More generally, worker cooperatives behave as if they are concerned with employment as well as income (Burdín & Dean, 2012; Craig & Pencavel, 1993). In contrast with CFs, worker cooperatives have been shown to respond to demand shocks by adjusting pay before and more than employment in the plywood industry of the US Pacific Northwest (Craig & Pencavel, 1992), in Italy (Pencavel et al., 2006) and in Uruguay (Burdín & Dean, 2009). Recent work on Indian worker cooperatives shows that loss of revenue does not result in proportional loss of employment and pay (Lal, 2023). Our paper contributes to this literature by comparing worker cooperatives in France with conventional French firms. Using a matched employer–employee dataset covering the whole population of employer firms in France in 2005–2015, we examine whether responses to ordinary demand shocks differ and are more equitable when firms are owned and managed by their employees. We look at two issues which are rarely addressed in the literature: (i) the use of hours worked, as well as employment and wages, to adjust

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firm production; and (ii) the possible heterogeneity of these adjustment mechanisms for different categories of workers.

France offers an interesting setting for examining these questions empirically. The country has a long-established cooperative movement with a significant number of worker cooperatives distributed across most industries (Fakhfakh et al., 2023) as well as very rich data that allow a systematic national comparison between all cooperatives and CFs. At the same time, extensive employment protection legislation and industry-level collective bargaining coverage may make both wages and employment downwardly rigid in both types of firms in France (Haltiwanger et al., 2014), so that the labour-managed and conventional sectors may not show significantly different responses to demand shocks in that country.

The paper provides new evidence on the choices of French firms generally in terms of employment and pay adjustments in response to changes in demand for the 2005–2015 period. Rather than examining a large worldwide economic shock, we focus on the consequences of more ordinary economic fluctuations and look at the impact of firms' idiosyncratic demand shocks. In addition, we have information on the number of hours worked, so that we can investigate whether nominal employment rigidity is mitigated by hours adjustments in either or both types of firm—an area where research is scarce.

Our other main objective is to investigate whether adjustments made by firms in response to shocks are distributed more equitably across groups of employees in LMFs. A question little addressed in the literature on participatory firms' adjustment choices is the heterogeneity of treatment for different categories of workers. One exception is Kim et al. (2018) which shows that parity codetermination in German firms increases job insurance and wage flexibility for skilled workers, who sit on codetermination boards, but not for unskilled workers, who don't. Different groups of employees may have different interests, objectives and levels of information, as well as varying abilities to defend their interests at the firm level. Generally, members finding themselves in a minority, for example, associated with job segregation, or faced with other forms of marginalization or discrimination may not have the same level of 'voice' as others despite the formal one-member, one-vote rule. Available evidence suggests the effect of employee participation on equality at work is not unambiguously positive in worker-managed firms (Magne et al., 2023). A handful of studies have looked at workforce heterogeneity in worker cooperatives, primarily focusing on pay. The recent study by Reibstein and Hanson Schlachter (2023) finds no unexplained wage gaps by gender, race or immigrations status in worker cooperatives in the United States but does report disparities reflecting, in particular, occupational segregation and homophily. Clemente et al. (2012) and Magne (2017) report mixed findings regarding the comparative extent of apparent gender discrimination in cooperatives and CFs in Spain and France, respectively. We look at whether individual employee characteristics, including gender and skills group, affect the treatment of employees in response to demand shocks and examine possible differences between cooperatives with different proportions of members in the workforce.[1]

Our results show that despite the specific French institutional context, employment stability in worker cooperatives—SCOPs (*sociétés coopératives et participatives*)—is significantly greater than in CFs, whereas wages are significantly more flexible, in line with international evidence. We find some evidence that hours are used as an adjustment tool specific to SCOPs. However, adjustments seem to differ across categories of workers in SCOPs in ways consistent with majority decision rules and common prejudices: There are significant differences between women and men, and between employees in high- and low-skill jobs.

[1] Unfortunately, we don't have data on race or ethnic background, which may not be collected under French law.

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In the next section, we outline the main characteristics of SCOPs, the theoretical framework and the hypotheses to be tested empirically for French worker cooperatives or SCOPs. The third section presents our data and empirical strategy, and the results and their interpretation are discussed in Section 4. Section 5 concludes.

2 | INSTITUTIONAL INFORMATION AND THEORY

2.1 | Worker cooperatives in France

There were 2700 SCOPs in 2023, employing a little more than 60,000 people (CG Scop 2024). SCOPs are for-profit LMFs owned primarily by their employees (worker members) partly with individual shares and partly collectively.[2] Members elect management and vote at least once a year on strategic issues. Each member only holds one vote regardless of the amount of capital they own. All employees may apply for membership, which is approved by the members' general meeting, typically after a probation period. The average proportion of members among SCOP employees is high, of the order of 75% (see Fakhfakh et al. (2023)). SCOP members are regarded as employees by French law and covered by the same provisions of labour law as their non-member colleagues (including for dismissals and quits, with imply loss of membership).

SCOPs are larger on average than other French firms and present in most industries,[3] though both groups comprise primarily small- and medium-sized firms. SCOPs are found more often in manufacturing and construction and in high-tech and IT services than other French firms. The distribution of their levels of capital intensity, industry by industry, is not significantly different from that of CFs (Fakhfakh et al., 2023). Finally, SCOPs are mandated by law to retain at least 16% of profit (the rule is 5% for CFs) but in practice retain more than 40%. The remainder is distributed to all employees (members and non-members) in the form of profit sharing (at least 25%, and in practice 46% in 2024—see CG-Scop 2025) and to all members as capital remuneration (up to 33% of profit). The rules on profit sharing provide incentives for existing members to increase membership among the firm's workforce if the SCOP needs the capital provided by new members as they join.

2.2 | Theory

In the traditional Ward–Domar–Vanek model (Domar, 1966; Vanek, 1970; Ward, 1958), LMFs maximize income per member and will have less elastic output and employment responses to changes in demand than profit-maximizing firms with the same technology satisfying conventional assumptions. With one variable input, the model produces the infamous 'perverse supply response' that sees LMFs respond to increases in product price with cuts in output and employment. However, as is well known, this result need not obtain with less stringent assumptions, and Pencavel and Craig's (1994) empirical findings do not support the hypothesis.

The lower elasticity of LMFs' employment responses to shocks than CFs' has been supported by evidence in the context of a different view of LMFs' objectives. Worker cooperatives have actually

[2] SCOPs may also have some external shareholders within strictly regulated limits on their share of capital and voting rights. The French legislation about SCOPs is reviewed in Fakhfakh et al. (2023).

[3] The exceptions are agriculture, financial services, real estate and public administration.

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been shown to behave as if employment as well as income, and indeed hours worked, matter to their members (Burdín & Dean, 2012; Craig & Pencavel, 1993). With pay endogenous, an LMF that pursues the collective welfare of its members is able to adjust pay as well as employment and hours worked.

When members value both wages and employment, it may be efficient to reduce wages instead of employment in case of downward economic shock, taking employees' preferences over employment and pay into account. However, this arrangement is complicated in CFs by information asymmetry between employers and employees (Aoki, 1984) and incentive incompatibility. Workers may not accept temporary wage reductions in order to secure their employment, because they do not know whether the cuts are justified, and employers will have incentives to cut jobs anyway, as well as keeping wages low even when the shock is over. Employers also do not necessarily know employees' preferences and have no guarantee that employees will not adjust their efforts downwards with wage reductions. As a result, firms typically do not adjust wages, which may be suboptimal. In contrast, in the LMF, worker members control the firm, and workers' preferences are known.

Recent empirical work on firms' responses to the Great Recession highlights workers' preference for employment stability when information is shared. This empirical literature suggests that whenever information asymmetry is lower (in the case of family-owned firms or employees' representatives on the board of directors), wages tend to be more flexible and employment more stable (Bassanini et al., 2013; D'Aurizio & Romano, 2013; Ellul et al., 2018; Gregorič & Rapp, 2019). Similarly, employee ownership in the form of broad-based ESOPs was also associated with greater employment stability in the United States (Kurtulus & Kruse, 2018).

Outside the specific context of the 2008–2010 recession, several papers compare the responses of LMFs to changes in input and output prices and/or demand shocks with the responses of CFs. Craig and Pencavel (1992; 1993) find that the plywood cooperatives of the US Pacific Northwest adjusted pay in response to changes in output prices in the 1970s and 1980s and had more stable employment levels than CFs in the same industry, which adjusted employment and hours.[4] With an extensive dataset, Pencavel et al. (2006) show that Italian worker cooperatives in the 1980s favoured employment stability while allowing more wage flexibility in response to transitory product market shocks than comparable capital-managed firms. Exploring the case of Uruguayan firms, Burdín and Dean (2009) again show LMFs have more stable employment and flexible wages for worker-members. Uruguayan LMFs also have job creation and destruction rates about half as high as those of capital-managed firms (Alves et al., 2016). Most recently, García-Louzao (2021) found that employment in Spanish worker cooperatives is more stable than in other Spanish firms. However, there is no comparative study of these issues on France, which has the third highest number of LMFs in Europe (after Italy and Spain).

Any substantial expansion or job cuts would likely have to be ratified by members in LMFs, and managers would still be accountable to members for smaller changes. Members are likely to resist downward adjustments of employment if those making the decision are at risk of being dismissed themselves (Bonin, 1981; Brewer & Browning, 1982; Steinherr & Thisse, 1979) or not re-elected in the case of managers. In addition, dismissals could conflict with cooperative members' values or objectives, which may include egalitarianism (Ireland and Law 1982) and/or the interests of co-workers (Sen, 1966). In the case of a negative demand shock, we therefore expect managers to be less willing to propose or make redundancies in SCOPs than in CFs. Finally, in a context of

[4] Berman and Berman (1989) also find US plywood cooperatives to have a more inelastic supply than their conventional counterparts.

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lower information asymmetry, they are able to adjust pay and hours worked to reflect employees' preferences better than CFs' managers (Berman & Berman, 1989).

Greater employment stability may also be observed in the case of a positive demand shock. Recruitment may be more limited in SCOPs in recoveries if job cuts have been more limited and the priority is to restore pay levels. SCOPs may also restrict employment expansion in anticipation of difficult dismissals. In addition, members may prefer not to expand the workforce since the potential cost of collective decision-making increases with size. Existing members may be reluctant to see their decision power diluted over time as new employees eventually join.

French CFs may also have had relatively low downward employment flexibility during the period of interest (2000–2015) (Haltiwanger et al., 2014). The OECD's 'indicator of employment protection' was 3.14 in 2013 for France, compared to 2.3 on average for OECD countries (OECD, 2013). Another factor that could increase employment stability in response to demand shocks in French CFs is the widespread presence of regulated profit-sharing schemes. In both statutory and voluntary regulated profit-sharing schemes, the share of profit to be distributed to employees is computed according to pre-set formulas in France, just as in the system proposed by Weitzman (1984). With such schemes, comparatively greater employment stability may come from more limited job cuts in response to negative shocks, as the firm only regards the fixed part of pay as the marginal cost of labour and has a higher demand for labour at a given level of total pay. As a result, fewer hires would also be needed in recoveries (Kruse, 1993: 122).[5] However, the limited evidence available to date suggests French CFs may regard profit-sharing bonuses as an integral part of the marginal cost of labour (Fakhfakh et al., 2013).

The first hypothesis we test is the following:

Hypothesis 1. Employment is less flexible upwards and downwards in SCOPs than in CFs.

SCOP members may prefer to adjust pay rather than employment in case of demand variation. Accepting a pay reduction during a downturn is made easier for SCOP workers by the assurance of being able to decide to readjust pay upward when the economic situation changes. As a result, pay may also be more variable upwards in SCOPs than in CFs.

Downward wage variations in France are limited by a national minimum wage and binding industry-level collective bargaining agreements, which apply to SCOPs. This degree of regulation probably is more stringent for worker cooperatives in France than regulations in the countries covered by existing empirical studies on the comparative response of LMFs and CFs to demand shocks.[6] This may cause a less marked difference in the elasticity of cooperatives' and CFs' pay responses in France than has been observed elsewhere if SCOPs have less scope for adjusting pay downwards.

We test the following hypothesis:

Hypothesis 2. Pay is more flexible upwards and downwards in SCOPs than in CFs.

[5] As noted by Nuti (1987), pay flexibility does not cause employment stability under profit-sharing in conventional firms, as the marginal cost of labour, that is, the wage, or fixed part of pay, is no more flexible under profit-sharing than without. This does not apply to worker cooperatives, despite the larger share of profit sharing in their pay, because pay is a choice variable for LMFs.

[6] Even in Italy where, as in France, collective bargaining coverage is exceptionally high (Izquierdo et al., 2017), those agreements were not binding on all firms in the period (1982–1994) covered by the data used by Pencavel et al (2006) depending on whether the employer belonged to a signatory organization of each particular agreement (De Matteis et al., 2011).

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Little evidence is available on the way hours worked respond, if at all, to changes in demand in all firms, due to the scarcity of data. Hours may be adjusted by LMFs as well as pay in response to demand shocks since hours are likely to enter the objectives of the LMF as well as pay and employment (Craig & Pencavel, 1993). Pencavel's (2014) empirical analysis shows that workers in plywood cooperatives in the Pacific Northwest in the United States behave like self-employed workers: They prefer to work more hours than are selected by management in CFs of the same industry, and their labour supply is a negative function of their hourly wage. In response to changes in output prices or demand shocks, Pencavel and Craig (1992) found that those same plywood cooperatives adjusted hourly wages more than employment or hours, whereas CFs adjusted worker-hours. However, hours worked could also be more flexible in SCOPs than in other firms as a substitute for job cuts, for example, if hourly pay cuts are limited by law (e.g., if workers are paid the minimum wage) and if members are more willing to adjust hours than CF employees within the legal limits. Hours worked are also regulated in France. The regulations have changed a number of times over the period covered by our data (Askenazy, 2013; Calavrezo & Lodin, 2012; ILO, 2012) but typically allow a certain amount of leeway for temporary adjustments in response to changes in demand. The same regulations apply to SCOPs as to CFs[7]:

Hypothesis 3. The number of hours worked is more elastic to demand shocks upward and downward in SCOPs than in CFs.

In practice, most LMFs throughout the world include both worker-members and workers who are not members. If the cooperative's objective is the collective welfare of its members, the hiring of non-member employees could lead to a dual workforce within firms, with members insured against the employment risk while non-members serve as an adjustment variable. Differences between members and non-members have been investigated theoretically (by Ben-ner, 1984; Kamshad, 1997 and Miyazaki, 1984) and empirically by Burdín and Dean (2009). Burdín and Dean (2009) find that in Uruguay, non-members' employment is no more likely to be adjusted than members'. SCOPs do have non-member employees, though French law provides incentives to increase membership among their workforce. On a 40-SCOP sample in the Rhône-Alpes region, Magne (2016) shows that most SCOPs do encourage membership for all workers after a 1-or-2-year trial period. Nonetheless, when deciding on job cuts or other adjustments, and if it comes to choosing which jobs will be cut, membership may play a role. Non-members, who don't have a vote, are more likely to be on probation and may be more likely to lose their jobs than members. Although we do not have information on individual employees' membership status, we do know what percentage of each SCOP's workforce are members. This measure depends not just on members' willingness to expand membership, but also on employment variation, as new employees typically work in the SCOP for a couple of years before becoming members. However, the measure allows us to present illustrative evidence on our fourth hypothesis, which follows:

Hypothesis 4. Employment will be even more stable in SCOPs with a high membership ratio. Low membership ratio SCOPs are expected to exhibit lower employment stability than high membership ratio SCOPs but higher than CFs.

[7] Our empirical approach allows us to control for specific provisions introduced in 2008 and 2009 to facilitate short-time working.

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Downward employment adjustment is dependent on the selection process of dismissed employees, which may not be the same in LMFs and other firms. Any employment expansion may also be affected by members' wish to hire employees with familiar characteristics. Member employees' involvement in employment decisions, whether directly or indirectly, implies that the characteristics of newly hired employees and of those made redundant in LMFs may significantly differ from other firms' new hires and redundancies as they reflect the preferences of the median member. Estrin (1991) shows that Italian worker cooperatives in manufacturing organize the production process in order to benefit blue-collar workers. Burdín (2016) finds a higher turnover for workers commanding the highest salaries in worker cooperatives compared with other Uruguayan firms. Lal (2023) finds that a large external shock on revenues results in significant negative effects on non-supervisory workers' employment and supervisors' wages. In the only existing in-depth investigation of workforce diversity dynamics, Dean (2022) shows the question of workforce homogeneity versus heterogeneity may not be the key issue for cooperatives hypothesized by Hansmann (1996). Dean (2022) finds there is no general pattern to those dynamics (in terms of membership status, skill level, gender and age) in LMFs in Uruguay. Worker cooperatives do have a more homogeneous workforce than other Uruguayan firms, especially among members, but whether the cooperatives evolve towards greater homogeneity, as implied by Hansmann's (1996) hypothesis, depends on the firm's cohort and industry (Dean, 2022). The ways different groups of members are included in collective democratic decisions within the firm are likely to affect their representation among redundancies and new hires.

In principle, equal voting rights and a generally more egalitarian outlook would suggest that, at least among members, all groups of employees, including often discriminated groups, would be better placed to defend their interests in an LMF than in a CF. However, majority rule and ordinary prejudices may act in the opposite direction. Based on the median voter theory (Kremer, 1997), we can form hypotheses regarding how well the interests of different categories of workers will be taken into account in the welfare function that is optimized by the LMF. Non-members, who do not vote at the general meeting, are likely to be disadvantaged, but so are minority groups represented at the general meeting. In SCOPs, minority groups include women (who only constitute 32% of the SCOP workforce—see CG-Scop 2025). Following Kremer (1997), the general assembly might not vote in favour of promoting equality between women and men. Indeed, it has been shown empirically that the gender wage gap is higher in SCOPs (Magne, 2017). More generally, prejudices may lead members (both men and women) to underestimate the skills, experience and contributions of employees from discriminated groups relative to those of non-discriminated groups and to hear their voices less. Prejudices are present among decision-makers in CFs as well, but given the minority status of women in collective decisions in SCOPs, women could actually be more likely to be made redundant (and less likely to be hired), as compared with men, in SCOPs than in other firms:

Hypothesis 5. Employment flexibility is lower in SCOPs than in CFs for women and men, but the difference between the two types of firm is smaller for women than for men.

We also look at possible differences between workers in high- and low-skill jobs.[8] High-skill workers' employment is likely to be less flexible than employment in other jobs in both SCOPs and

[8] An argument can also be made for recently hired employees being made redundant first, for example, if only because they are more likely to be on probation. However, the position of new hires is substantially influenced by other issues like

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CFs due to their rarer skills that protect them from competition on the labour market. However, we expect this difference between high- and low-skill jobs to be narrower in SCOPs because workers with lower skill levels are more involved in decision-making in LMFs than in CFs and because high-skill workers constitute a minority:

Hypothesis 6. Employment flexibility is lower in SCOPs than in CFs for low-skill workers and high-skill workers, but the difference between the two types of firm is smaller for high-skill workers.

3 | DATASET AND EMPIRICAL STRATEGY

3.1 | Data

In order to test our six hypotheses, we merge two datasets: the matched employee–employer DADS and the FICUS-FARE. The DADS provide individual employees' pay and hours information collected from all French firms every year by the National Institute of Statistics (INSEE) and made available for researchers via a secure procedure. Its completeness is guaranteed by the fact that it is a compulsory report that every employer must make every year. We use firm-level observations from 2005 to 2015. We are interested in three main variables: The number of employees, pay and the total number of hours worked for a given year in a given firm. In addition, we have a number of control variables: industry, location, age of the firm and precise composition of the workforce (proportion of women, employees in high-skill jobs,[9] full-time workers and types of contract).

The FICUS-FARE dataset includes financial variables for all firms with employees. From this dataset, we build a proxy variable for the demand shock that applies to a firm during a given year (the details of this variable construction are given in the next section) and obtain information about the firm's capital. The two datasets—DADS and FICUS-FARE—include firm identifiers that make matching possible. In order to measure the impact of demand variations on wages and hours, we also use the individual employee-level observations of the DADS.

Some observations are eliminated industries without SCOPs[10]; households employing personal-service workers; the public sector; non-profit organizations, temporary employment agencies and all firms with no sales in the year considered. All monetary values are deflated using the consumer price index. To eliminate reliance on extreme outliers, all variables are winsorized at the 1% level. We obtain a panel dataset of 7.3 million observations for 2005–2015 at the firm level and a dataset of 15 million observations at the individual employee level.

their skill level, and empirically, their presence is likely to be correlated with employment expansion. We briefly discuss below the inconclusive results we obtained regarding length of service (see Section 4.4).

[9] Six categories of occupations are identified in the data: managers and professionals, intermediate occupation, semi-skilled white-collar, unskilled white-collar, semi-skilled blue-collar, unskilled blue-collar (see Table A1). When splitting our sample between workers in 'high-skill' and 'low-skill' jobs, we include managers and professionals as well as intermediate occupations in the first category and all clerical and manual workers in the second one. For robustness, we also do all regressions with an alternative cut-off point: managers and professionals on one hand and all other categories on the other hand. All our results hold.

[10] Specifically: Agriculture, finance and insurance, real estate and public administration.

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Because our dataset is very large, we are able to construct matched samples in order to reduce confounding bias and model dependency (Iacus et al., 2012). We build two matched samples, using coarsened exact matching as it requires fewer assumptions than other matching methods and is particularly well suited for large control groups (Blackwell et al., 2010).[11]

3.1.1 | Firm-level observations matching

Using Coarsened-Exact Matching, we find exact matches for each worker cooperative using four variables: industry at the two-digit level, the exact age of the firm (in years), the number of workplaces of the firm and level of capital (six categories[12]). We do not match on the number of employees in the firm as this is our main outcome variable. The region is also excluded from the matching variables because a significant proportion of firms have several workplaces located in different regions. Because many firms are only in operation for a subset of the years in our sample, we are left with a slightly unbalanced sample of 32,979 firm-year observations, 17,198 of which are worker cooperatives (5672 firms, among which 2836 are cooperatives).

3.1.2 | Individual-level observations matching strategy

For the employee-level equations, we match the workforce of a given SCOP with that of a similar CF using coarsened exact matching on a slightly different set of variables. In addition to industry and firm age, we now include the region and the number of workers (five categories). We then use all workers who worked in an SCOP or a matched CF at any one point during the 2002–2015 period.

3.2 | Descriptive statistics

Tables 1 and 2 show the means of our variables for CFs and cooperatives in all four samples: the unmatched firm-level sample, the unmatched individual-level sample and the two matched samples. As expected, the characteristics of cooperatives and CFs are more similar in the matched samples. On average, SCOPs are larger than CFs and offer higher wages and longer working hours. They are found in the construction and manufacturing industries more than CFs, and as a result the proportion of white collars is lower, but the proportion of semi-skilled blue-collars is much higher even in the sample matched by broad industry. Similarly, the proportion of women is markedly lower in SCOPs, even in the matched sample. Cooperative workers also tend to be older and have longer tenure.

Figures 1–3 show the evolution of average number of workers, the total number of hours worked per worker and the average hourly gross pay in the two groups of firms, using the unmatched samples to have a first, general picture of the apparent dynamics in the two types of firms. A few

[11] According to Blackwell et al. (2010), the problem with Mahalanobis distance or propensity score matching is that it requires the user to set the size of the matching solution ex ante and then check for balance ex post. Coarsened exact matching, on the other hand, may reduce more considerably the number of observations. However, this is not a problem in our case as the untreated group (conventional firms) includes several million observations.

[12] Details about the thresholds can be found in Supporting Information Appendix 2.

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TABLE 1 Variable means—firm level.

	Unmatched sample			Matched sample (with non-switching cooperatives)	
Variables	**Conventional firms**	**Non-switching cooperatives**	**Switching cooperatives**	**Conventional firms**	**Cooperatives**
Number of workers Mean	9.9	14.3	17.8	12.6	14.0
Average pay	15.02	16.52	16.43	16.22	16.45
Average capital (€ thousands)	873.04	890.66	849.03	803.87	760.52
Firm age (years)	13.1	15.4	17.5	15.3	14.8
Manufacturing industry (%)	12.0	20.1	14.8	19.5	19.5
Construction industry (%)	15.5	26.3	23.0	27.9	27.5
Service industry (%)	72.47	53.6	62.2	52.6	53.0
Number of firms	1,380,508	2883	121	2738	2738
Number of firm-year observations	7,718,197	17,547	1070	15,214	16,378

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TABLE 2 Variable means—employee level.

Variables	Unmatched sample		Matched sample	
	Conventional firms	Cooperatives	Conventional firms	Cooperatives
Mean of real hourly pay	17.0	16.34	15.44	16.34
Proportion of women (%)	39.5	27.1	37.2	27.1
Managers and professionals (%)	12.9	11.2	11.5	11.2
Intermediate occupation (%)	16.9	19.0	16.0	19.0
Semi-skilled white-collar (%)	9.9	8.1	9.1	8.1
Unskilled white-collar (%)	18.8	4.9	17.7	4.9
Semi-skilled blue-collar (%)	24.3	37.0	25.3	37.0
Unskilled blue-collar (%)	13.5	16.7	17.0	16.7
Average age	37.2	40.2	35.4	40.2
Actual hours worked within a year	1290	1362	1198	1362
Tenure (years)	3.8	5.5	2.7	5.5
Number of individuals	2,752,067	6468	23,947	6468
Number of year-individual observations	15,679,510	44,346	171,550	44,346

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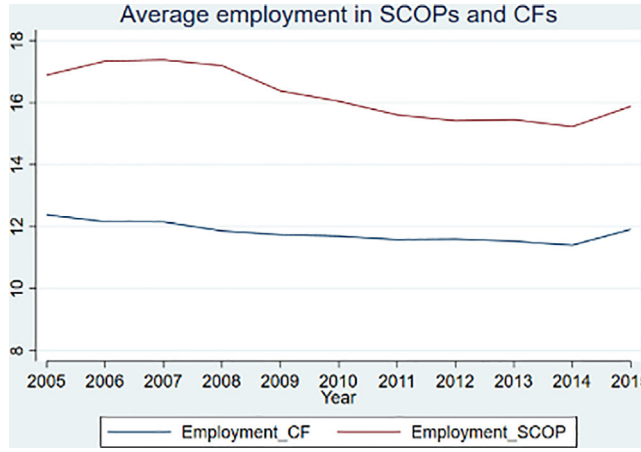


FIGURE 1 Evolution of the average employment in *sociétés coopératives et participatives* (SCOPs) and conventional firms (CFs). *Note*: Average employment is calculated as the average of the number of employees in all firms at the end of the year. [Colour figure can be viewed at wileyonlinelibrary.com]

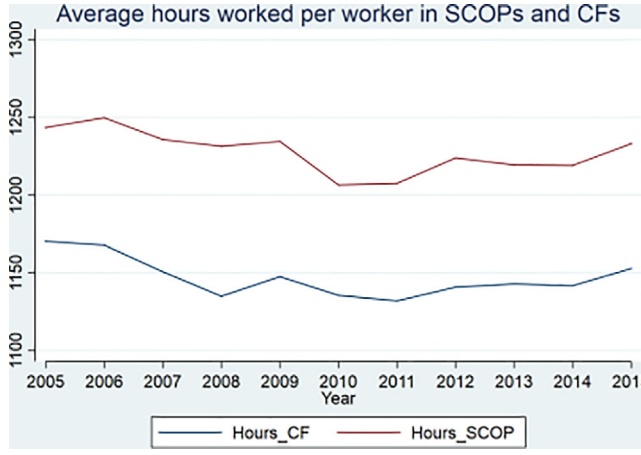


FIGURE 2 Evolution of hours worked per worker in *sociétés coopératives et participatives* (SCOPs) and conventional firms (CFs). *Note*: The number of hours is calculated yearly for all firms. [Colour figure can be viewed at wileyonlinelibrary.com]

differences are worth noting. The number of workers, hours worked and hourly pay are all higher in SCOPs. Employment increases and decreases with market conditions and appears to be slightly more variable in SCOPs. The number of hours worked per worker in a year and hourly pay also seem more volatile in SCOPs, particularly in 2009 at the peak of the financial crisis.[13] Figures 4 and 5 show that job creation and job destruction rates over a year are both lower in SCOPs but seem to vary more with the economic cycle.

We are interested in the adjustment strategies of SCOPs and CFs when they are faced with the same demand shocks. It is sometimes argued that cooperatives' resilience could be due to their niche activities being protected from competition. This is not what we observe in Figure 6: If anything, sales in SCOPs seem more affected by economic cycles. Finally, the proportion of bankruptcies (Figure 7) is lower in SCOPs over the whole period but increases more sharply in 2009.[14]

[13] GDP growth in France was 0.26% in 2008, −2.87 in 2009 and 1.95 in 2010 (World Bank national accounts data).

[14] Figures 1–7 are also analysed for the matched sample (see Supporting Information Appendix A3). The trends remain similar although the patterns for matched conventional firms are closer to the cooperatives on average.

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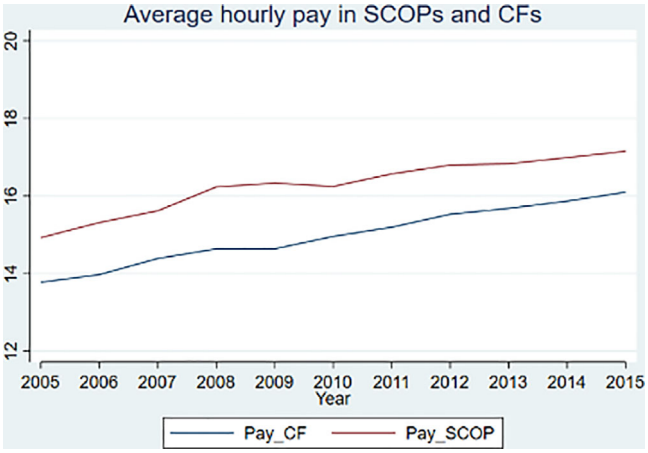


FIGURE 3 Evolution of average hourly pay in *sociétés coopératives et participatives* (SCOPs) and conventional firms (CFs). *Note*: Pay includes all profit-sharing, bonuses and social insurance contribution. [Colour figure can be viewed at wileyonlinelibrary.com]

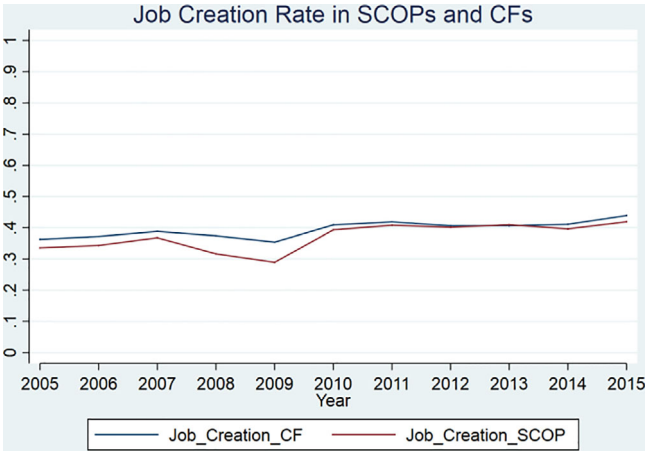


FIGURE 4 Job creation rate in *sociétés coopératives et participatives* (SCOPs) and conventional firms (CFs). *Note*: Job creation rate is calculated every year as the total number of job creations divided by the total number of jobs. [Colour figure can be viewed at wileyonlinelibrary.com]

3.3 | Empirical strategy

Our empirical strategy starts from the existing international empirical literature on the subject. Empirical studies find worker cooperatives to have more flexible wages and more stable employment than CFs (Craig & Pencavel, 1993; Pencavel et al., 2006; Burdín & Dean, 2009). Based on these reference articles, we estimate three equations in reduced form, explaining employment, the average pay (including profit-sharing[15]) and the number of hours effectively worked (including overtime), in log-linear form. Existing studies on the issue of responses to demand shocks analyse the firm level. Our dataset allows us to add analyses at the level of the individual employee. Thus, our three equations are estimated once at the firm level and once at the level of the individual employee (explaining the probability of leaving the firm, pay and hours). In what follows, we discuss specification and estimation issues first for the linear estimations for the employment equation at the firm level and the pay and hours equations (both firm- and employee levels); then the estimation of hazard functions for the employment equation at the employee level; and finally the specific issues involved in specifying the demand shock variable. Variable definitions appear in Supporting Information Appendix A1.

[15] Unfortunately, it is not possible in the data to disaggregate pay into its fixed and variable components.

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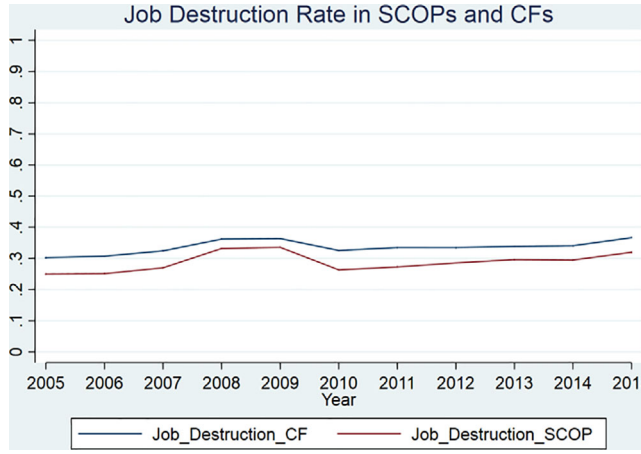


FIGURE 5 Job destruction rate in *sociétés coopératives et participatives* (SCOPs) and conventional firms (CFs). *Note*: Job destruction rate is calculated every year as the total number of job destructions divided by the total number of jobs. [Colour figure can be viewed at wileyonlinelibrary.com]

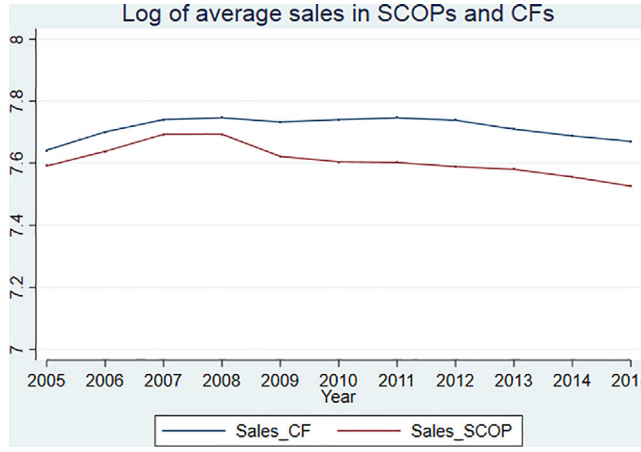


FIGURE 6 Evolution of average sales in *sociétés coopératives et participatives* (SCOPs) and conventional firms (CFs). *Note*: Log of average sales is measured as the log of average sales for all SCOPs in a given year and for all CFs in a given year. [Colour figure can be viewed at wileyonlinelibrary.com]

Our overall linear specification for the employment, pay and hours equations is the following:

$$\ln Y_{it} = \alpha_0 + \alpha_1 \text{SCOP}_{it} + \alpha_2 D_{it} + \alpha_3 \text{SCOP}_{it} D_{it} + \alpha_4 Z_{it} + \alpha_5 \text{SCOP}_{it} Z_{it} + \alpha_6 X_{it} + \beta_n \text{YearDummy_}n + \sigma_i + \varepsilon_{it} \quad (1)$$

where $\ln Y_{it}$ is the outcome variable representing first employment at the firm level, second pay at the firm level and at the individual level and third actual hours worked at the firm level and at the individual level (for employment at the individual level, we estimate hazard functions—see below). D_{it} is the proxy variable used to measure idiosyncratic demand shocks faced by the firm (details of the calculation below). SCOP_{it} is a dummy variable equal to 1 if firm i is an SCOP in year t and 0 if it is a CF. Z_{it} is a vector of variables including the logarithm of the capital stock of firm i in year t and, in the employment and hours equations, the logarithm of total pay. Interaction terms between the SCOP dummy and other explanatory variables are included in the regression to measure any significant difference between SCOPs' and CFs' responses.[16] X_{it} is a vector of

[16] As shown by Fakhfakh et al. (2012), SCOPs and CFs may have different production functions. These interaction terms in our regression take these into account.

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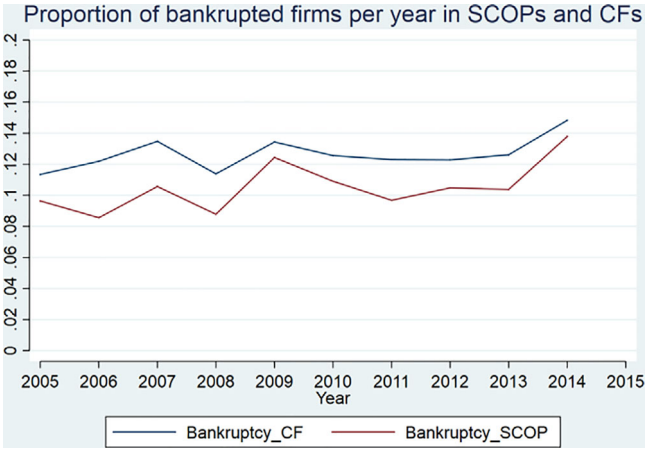


FIGURE 7 Bankrupcy rate in *sociétés coopératives et participatives* (SCOPs) and conventional firms (CFs) per year. *Note*: The proportion of bankruptcies is calculated as the number of firms that are not in the data the following year divided by the total number of firms on a given year. These first descriptive statistics provide support to hypotheses H2 and H3 about wage and hours flexibility but not to H1 about employment adjustments. [Colour figure can be viewed at wileyonlinelibrary.com]

relevant variables that differ across the equations estimated. For the firm level, this vector includes firm age; workforce composition variables (proportion of women, proportion of managers and professionals, intermediate occupations, semi-skilled white-collars, unskilled white-collars, semi-skilled blue-collars, unskilled blue-collars, average employee age and age square) and regional unemployment rate. At the individual level, X_{it} includes occupation dummies (six categories as above), age and age squared, tenure and firm size. We also include year dummies to control for macroeconomic fluctuations that affect all firms. We use clustered standard errors at the firm level or at the individual level.

3.3.1 | Employment—firm-level equation

For the employment equation at the firm level, $\ln Y_{it}$ is the logarithm of the number of workers in firm i on December 31st of year t, and the vector Z_{it} includes the logarithm of average hourly pay for firm i in year t, including all work-related income.[17] For CFs, we expect employment to decrease with the price of labour and capital to have a positive impact on employment. For worker cooperatives, the sign of the relationship between pay and employment is ambiguous because pay is endogenous (Craig & Pencavel, 1992). If hypothesis H1 about employment stability in cooperatives is confirmed, we expect D, the demand shock variable, to have a positive coefficient and the interaction term with the SCOP dummy a negative coefficient.

Hypotheses H5 and H6 (about employment stability by gender and broad skill group, respectively) will also be tested by including interaction variables SCOP × gender, SCOP × high-skill job.

The employment equation raises a potential endogeneity problem as the level of pay and the level of employment are determined simultaneously. We use the second and third differences of the log of hourly pay to instrument the pay variable.[18]

Some French firms change status from CF to SCOP or (occasionally) from SCOP to CF. Over the whole period from 2005 to 2015, about 200 firms switched (150 from CF to SCOP and 50 from SCOP to CF). This characteristic was used for comparing productivity in converted CFs and firms

[17] The pay variable includes the fixed wage as well as flexible pay components and profit-sharing.

[18] Burdín and Dean (2009) solve this problem by instrumenting the wage variable by its lag in a very large panel with monthly observations. In our 10-year panel, this is not an option since the error term includes past values and will therefore be correlated with the lagged wage.

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who could have converted but did not (Young-Hyman et al., 2022). However, it is not suited for comparing the level of employment as very specific workforce changes typically occur due to the conversion, which are not generalizable to all cooperatives.[19] There are two types of conversions of CFs to SCOPs: rescue worker buyouts and buyouts of sound CFs (e.g., when the historical owner retires). In the case of rescue buyouts (about 25% of conversions), the new cooperative typically starts with a fraction of the pre-conversion workforce (usually between 30% and 70%) and is very unlikely to hire anyone in the next few years. In the case of sound firm conversions (about 75% of conversions), several departures often also happen at the same time, which can be significant for small firms. In order to find a good match for this second type of conversion, we would need to select CFs whose owner/manager retires at the same time, but unfortunately we do not have that information in the data. The fewer cases of conversions from SCOPs to CFs exclusively concern cases where the survival of the firm is in danger and often involve turbulent transitions and mass redundancies.[20] We eliminate these from our sample.

We run our main estimations removing the switching firms and including fixed effects but no SCOP dummy (since without the switching firms, it becomes included in the fixed effects). In estimations not reported here, we ran regressions including switching firms as a robustness test, and our results are robust to their inclusion.

3.3.2 | Pay

In order to test hypothesis H2, we estimate the impact of idiosyncratic demand shocks faced by the firm on hourly pay. The outcome variable in Equation (1) becomes the average hourly pay in firm i for year t in the firm-level equation and the gross hourly pay for individual j in year t in the employee-level equation.

There could also be endogeneity issues in this set of estimations, for example, if workers with characteristics that would yield them higher (or lower) wages tend to opt into cooperatives. We deal with potential endogeneity in the firm-level equation by matching each SCOP with a similar CF on region, firm size, industry and year of creation, using coarsened exact matching.[21] For the employee-level estimation, we also restrict the sample. First, we exclude workers who switch firm during this time period, because we want to capture the firms' internal pay policy rather than the effect of employee mobility on individual wages. Second, we only keep workers who work for the whole year, because in practice it often is the condition for being entitled to profit-sharing. We include all the employees who worked in an SCOP or a matched CF at any point during the 2005–2015 period.

3.3.3 | Hours

As firms may adjust hours rather than employee numbers, we test hypothesis H3 with two estimations, at the firm- and employee levels, respectively. For the firm level, the outcome in Equation (1)

[19] For the same reason, another possible approach, Difference in Differences, is not appropriate here.

[20] Leaving the SCOP status is permitted in exceptional circumstances if the survival of the firm is at stake and the change is authorised by the French federation of worker cooperatives, CGSCOP.

[21] Matching only reduces endogeneity if the variables influencing treatment assignment are observable but does not help with unobserved confounders. In our case, there is no obvious unobserved variable systematically influencing treatment assignment (being a cooperative).

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becomes the average number of hours worked per employee for firm i during year t. We use the same matching strategy as for the pay equations. For the employee level, the outcome is the total number of hours worked by individual j during year t. Once again here we exclude workers who switch firm during this time period or only work for part of the year, because we want to control for the impact of recruitment and dismissal of part-time workers.[22]

3.3.4 | Employee-level employment

At the employee level, we explain the individual probability of departure by estimating a hazard function. Following Burdín (2014) and Dean (2019), we estimate the hazard of an individual j working in firm i in year t leaving the firm the following year $(t + 1)$. In other words, we estimate the probability of leaving, subject to having been employed in firm i until then. We use a Cox proportional hazard model with delayed entry (Cox, 1972). This model presents several advantages for our purpose. First, it is semi-parametric and, as such, does not require specifying a baseline hazard, including restrictive distributional assumptions. Second, it allows delayed entries and censoring, fitting our data since some individuals in the sample entered after 2005, and some have started working in a firm before 2005 and/or keep working in a firm after 2015.[23] Third, it allows us to include time-dependant covariates. Finally, we are able to take into account multiple failure-time data. In our sample, approximately a quarter of the individuals have several employment periods in different firms during the 2005–2015 period. Failure times are therefore correlated within clusters, which violates the independence of failure time assumption required in traditional survival analysis (Cleves, 2000). We assume that departures are unordered events (they could occur in any sequence) and that they are recurrences of the same type of events (Therneau, 1997). We use clustered standard errors at the individual level. We estimate the following equation:

$$\lambda(t, Z) = \lambda_0(t) \exp[\beta Z(t) + \gamma \text{SCOP}] \tag{2}$$

where $\lambda_0(t)$ is the non-parametric estimate of the baseline hazard. Z_{jt} is a vector of individual characteristics, including gender, age, number of hours worked per day, position in the firm (high- or low-skill position) and firm characteristics including industry, size and the demand shock. This allows us to test hypotheses H1 and H4 (greater employment stability in SCOPs and even higher when membership ratio is high) while taking into account individual characteristics. We expect γ to be negative—controlling for the demand shock, the probability of leaving the firm in year t, given that the worker was still working in year $t-1$ is expected to be lower in an SCOP. In order to test our hypothesis H4 that SCOPs with higher membership rates have more stable employment, we estimate hazard functions for SCOPs with different proportions of members in their workforce.

It should be noted that Equation (1) concerns both downward and upward adjustments at the firm level, but the individual employee's hazard of leaving the firm in Equation (2) only measures downward employment adjustments. Another issue is that we are not able to differentiate between quits and redundancies. Quits and redundancies are both expected to be affected by demand shocks: In the case of a negative demand shock, redundancies are likely to be higher,

[22] For example, in the case of downward demand shock, firms can reduce the number of hours per worker, while simultaneously dismissing part-time workers (who are likely to be among the least secure jobs). This will conceal the use of hour reductions to adjust production.

[23] Meaning that we do not observe all times of entry and all times of failure (departure from the firm).

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but quits are likely to be lower as employees anticipate to be less likely to find a job in another firm. However, both are likely to be lower in SCOPs than in other firms if our hypotheses are correct.

3.3.5 | Specification of the demand shock variable

One key empirical problem is to find a suitable proxy variable to measure the demand shock that applies to firms for a specific year. Pencavel et al. (2006) use the deviation of the firm's sales in a year from the sales predicted by a linear regression. Burdín and Dean (2009) use the yearly variation in the firm's sector output price for each month and sector. We do not have output prices. We build two variables as follows.

Deviation of the firm sales from their linear trend

The first proxy variable is the deviation of the firm's sales from the sales predicted by a linear regression. It is directly inspired by Pencavel et al. (2006). We regress the logarithm of firms' sales on a time trend and firm fixed effects:

$$\ln_\text{Sales}_{it} = \beta T + \mu_i + \varepsilon_{it}$$

where $\ln_\text{Sales}_{it}$ is the logarithm of the sales of firm i for year t and T is a continuous time trend. The estimated coefficient β represents the average linear trend of sales evolution for all firms. μ_i measures firm-specific fixed effects, and ε_{it} is the error term. Our proxy variable for the demand shock is $D_{it} = \mu_i + \varepsilon_{it}$.

This variable is particularly relevant because one possibility for explaining more stable employment in cooperatives could be that they operate in niche activities and are less affected by demand shocks than CFs. By using a firm's idiosyncratic demand shock, we can measure what firms do when similarly confronted with individual shocks, therefore isolating the effect of adjustment choices.

However, there is a risk of endogeneity with the employment variable as employment and sales are determined simultaneously. Measuring the deviation from the trend eliminates part, but not all, of the endogeneity compared to the use of gross sales. As a result, we might over-estimate coefficient α_1 in Equation (1) because it will not only capture the impact of the sales' deviation on employment but also in part the effect of employment on the deviation.

Lag of deviation of the firm sales

Using contemporaneous sales deviations risks simultaneity bias as demand shocks and employment may be jointly determined so. To avoid this bias, we lag D_{it} by one period.

Deviation of the industry sales from the industry linear trend

This third proxy variable aims at overcoming the endogeneity problem by taking into account the deviation of industry sales from its trend. We estimate the following equation:

$$\ln_\text{SalesIndustry}_{kt} = \alpha T + \theta_k + \vartheta_{kt}$$

where $\ln_\text{SalesIndustry}_{kt}$ is the sum of sales for all firms from industry k (at the four digit level) for year t, and T is a linear time trend. Our proxy variable is then $D'_{it} = \vartheta_{kt}$.

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4 | ECONOMETRIC RESULTS AND INTERPRETATION

4.1 | Employment

Table 3 presents the results of our estimation of Equation (1) for the matched sample, estimated by instrumental variables with fixed effects (Column 1) as well as without instrumentation (Column 2). Columns (3)–(6) show estimations with and without IV for the unmatched sample, first for firms that don't switch from conventional to SCOP (Columns 3 and 4) and then for those that do (Columns 5 and 6) for comparison. Estimated coefficients are all of the same order of magnitude in the IV estimations, confirming our valid instruments are not weak. Similarly, their overall stability across samples with the same estimation methods suggests that there is no strong bias induced by any endogeneity of switching to or from the SCOP form. All the coefficients have the expected signs for all firms. Overall, idiosyncratic demand shocks have a positive impact on employment—when sales are higher than their trend for the whole period employment increases and when they are lower, employment decreases. Pay has a negative impact on employment, as predicted by economic theory. Similarly, capital has a positive impact on employment.

The coefficient associated with the demand shock interacted with the SCOP dummy (α_3 in Equation (1)) is negative: employment is found to be significantly less elastic to demand shocks in SCOPs than in CFs, verifying hypothesis H1.[24] When sales are 10% lower than the trend, firms lower employment by 7.8% on average, but employment is cut by only 6.2% in SCOPs. This result is illustrated by computed margin plots (see Supporting Information Appendix A4). Hypothesis H1 was about employment stability in response to demand shocks, whether they are positive or negative. However, employment responses to positive and negative shocks may differ. Regressions using the alternative demand shocks are reported in Supporting Information Appendixes A5 and A6 and pooled OLS in Supporting Information Appendix A7. They all give similar results. We ran regressions separating positive and negative shocks by adding a dummy variable for positive/negative shocks interacted with the demand shock and then with the cooperative dummy. Results are reported in Supporting Information Appendix A8 and show some evidence that employment stability is mostly driven by downward stickiness.

Given that the sample period includes the 2009–2010 recession, this systemic shock could dominate the estimated responses. In order to test this, we ran the same regressions, restricting our sample and recalculating to the 2011–2015 period, and found that the results hold in normal times (see Supporting Information Appendix A9).

One result from existing studies is not confirmed by our regressions. As pay is endogenous in LMFs, the sign of the relationship between pay and employment is in principle indeterminate in LMFs (Craig & Pencavel, 1993) and has been found to be significantly different in LMFs and CFs (Pencavel et al., 2006; Burdín & Dean, 2009). However, we do not observe any significant difference between SCOPs and other firms in France in that respect. This finding is consistent with García-Louzao's (2021) result for Spain, which he attributes to labour market regulation. The national minimum wage and exceptionally large coverage of binding industry-level collective

[25] The SCOP dummy is relevant only when switching firms is included (otherwise, it is part of the firm fixed effects). It reflects CFs that switched to SCOPs.

[24] This means that the slightly higher variability that appears in Figure 1 for SCOPs is due to higher exposure to demand shocks for SCOPs overall. However, when SCOPs and CFs with similar characteristics are exposed to the same demand shock, SCOPs adjust employment less.

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TABLE 3 Employment equation, firm level (pay instrumented with lags 2 and 3).

	(1)	(2)	(3)	(4)	(5)	(6)
	Matched sample		**Whole unmatched sample non-switchers**		**Whole unmatched sample switchers**	
	Estimation method: 2SLS	**Estimation method: OLS**	**Estimation method: 2SLS**	**Estimation method: OLS**	**Estimation method: 2SLS**	**Estimation method: OLS**
	Employment (ln)	Employment (ln)	Employment (ln)	Employment (ln)	Employment (ln)	Employment (ln)
Demand shock	0.857*** (0.03)	0.808*** (0.04)	0.805*** (0.00)	0.734*** (0.00)	0.805*** (0.00)	0.734*** (0.00)
SCOP × demand shock	−0.161*** (0.04)	−0.168*** (0.04)	−0.190*** (0.03)	−0.157*** (0.02)	−0.154*** (0.02)	−0.142*** (0.02)
Pay	−0.281 (0.20)	−0.293*** (0.07)	−0.407*** (0.01)	−0.258*** (0.00)	−0.407*** (0.01)	−0.258*** (0.00)
Capital	0.146*** (0.02)	0.115*** (0.02)	0.184*** (0.00)	0.163*** (0.00)	0.184*** (0.00)	0.163*** (0.00)
SCOP[25]					0.635 (0.69)	0.363+ (0.20)
SCOP × pay	0.064 (0.33)	0.051 (0.09)	0.058 (0.28)	0.050 (0.07)	−0.157 (0.26)	0.047 (0.06)
SCOP × capital	−0.002 (0.03)	−0.058* (0.03)	0.003 (0.02)	−0.035+ (0.02)	0.007 (0.02)	−0.030+ (0.02)
Firm−fixed effects	YES	YES	YES	YES	YES	YES
2SLS, pay instrumented with second and third lags	YES	NO	YES	NO	YES	NO
N	16,576	30,470	3,714,134	7,293,067	3,715,448	7,295,242
Including number of SCOP obs	9076	16,059	9739	17,200	10,498	18,262
Sargan overidentification test	5.564 *p* value = 0.0619		5.004 *p* value = 0.0819		8.303 *p* value = 0.0563	

Note: Firm-clustered standard errors in parentheses. All models include the following controls: firm age, workforce composition variables (proportion of managers and professionals, intermediate occupation, semi-skilled white-collar, unskilled white-collar, semi-skilled blue-collar, unskilled blue-collar, average employee age and age square and proportion of women), regional unemployment rate and year dummies.

*** indicates 0.1% level of significance, **1%, *5% and +10%. All models include firm and year fixed effects.

Abbreviation: SCOP, *sociétés coopératives et participatives*.

$+p < 0.10$.

$^{*}p < 0.05$.

$^{**}p < 0.01$.

$^{***}p < 0.001$.

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TABLE 4 Hazard of leaving the firm for employees (Cox proportional hazard function).

	Hazard ratio for leaving the firm
Demand shock	0.802*** (0.02)
SCOP	0.746*** (0.01)
Pay	0.858*** (0.02)
High-skill job worker	0.966 (0.03)
Men	0.961*** (0.01)
N	144,223
Including number of SCOP obs	29,038

Note: Firm-clustered standard errors in parentheses. Included controls: occupation dummies (six categories: professionals, intermediate occupation, semi-skilled white-collar, unskilled white-collar, semi-skilled blue-collar and unskilled blue-collar), age and age squared, gender, seniority and firm size. Significance tests are based on null hypothesis that hazard ratio = 1.0. All regressions include controls for worker occupation, firm capitalization, size of the firm and number of hours worked, as well as age and seniority of the workers.

Abbreviation: SCOP, *sociétés coopératives et participatives*.

+$p < 0.10$.

*$p < 0.05$.

**$p < 0.01$.

***$p < 0.001$.

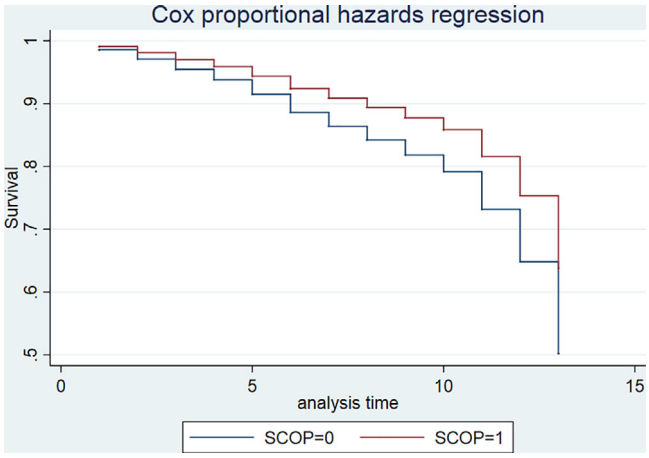


FIGURE 8 Estimated survival curves for *sociétés coopératives et participatives* (SCOPs) and CFs workers. *Note*: A log-rank test was run and shows that the two survival functions are significantly different for worker cooperatives and conventional firms. [Colour figure can be viewed at wileyonlinelibrary.com]

bargaining agreements in France, as well as the high level of employment protection there, may explain our result.

Results for the employee-level hazard function (2) in Table 4 confirm the findings from the firm level that downward employment flexibility is lower in SCOPs when controlling for demand shocks, validating hypothesis H1—workers in SCOPs are 25% (1–0.746) less likely to leave the firm in a given year than workers in CFs.

The survival function from the Cox proportional regression for SCOPs and CFs is represented in Figure 8. Every year, the hazard of leaving the firm is much lower in SCOPs than in CFs, so that the

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employee's probability of staying with the firm at a given time, subject to having been employed there until then, is higher in an SCOP, controlling for all our firm and individual characteristics.

4.2 | Wages

Given their greater employment stability, we expect greater pay variability in SCOPs than in other firms (hypothesis H3). Tables 5 and 6 show results for the wage equations at the firm- and employee levels, respectively. Again, the estimated coefficients are quite stable across subsamples and have the expected signs. Both sets of results confirm average hourly pay varies with demand shocks in all firms, possibly thanks to profit sharing (which is included in the pay variable) but is significantly more variable in SCOPs.

4.3 | Hours

Table 7 shows the estimation results explaining hours worked at the firm level. The dependent variable is the logarithm of the number of hours worked per worker in a firm during a year. The impact of a demand shock is positive: when sales are higher than their trend for the whole period, the number of hours worked increases. However, this positive impact is higher in SCOPs, and the difference is significant, which suggests that hours represent an additional choice for adjustment for SCOPs.

However, this interpretation is not really confirmed when we estimate the equation at the employee level (see Table 8), including only workers who stay in the firm for the whole year and do not switch firms, in order to avoid confounding effects associated with the use of part-time employees. The specific SCOP effect of the demand shock is only very weakly significant in the estimation using the unmatched sample. Thus, we find only limited evidence for hypothesis H3 that in the context of employment rigidity, SCOPs use hours as an adjustment variable.

Firms could also use part-time workers and full-time workers differently when adjusting hours worked. In our data, part-time workers represent about 25% of the workforce. In order to test this hypothesis of different use of part-time and full-time workers, we ran the regression again, excluding part-time workers. We found that in this new estimation, the SCOP coefficients are slightly more significant than for the whole sample (see Table 8, Column 3). This could suggest that SCOPs use full-time workers' hours to adjust for the total number of hours worked, potentially distributing the cost of adjustment more equitably, while CFs use mainly part-time workers.

4.4 | Employment responses for different groups of workers

Our estimates of the ratios between the hazard of leaving the firm in CFs and in SCOPs with different proportions of members among their employees confirm that the difference is greater, the higher the proportion of members among the cooperative's workforce (hypothesis H4—see Figure 9).[26] Low-membership ratio SCOPs exhibit lower employment stability than high-membership ratio SCOPs but greater stability than CFs as long as the membership ratio remains

[26] Our data do not allow us to test if non-members are more likely to be made redundant as the only information we have is the proportion of workers who are members at the firm level.

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TABLE 5 Pay equation at the firm level, firm fixed effects.

	(1) Matched sample	(2) Whole unmatched sample non-switchers	(3) Whole unmatched sample switchers
	Average hourly pay (ln)	Average hourly pay (ln)	Average hourly pay (ln)
Demand shock	0.042*** (0.01)	0.036*** (0.00)	0.036*** (0.00)
SCOP × demand shock	0.018* (0.01)	0.011** (0.01)	0.010** (0.01)
Capital	−0.007+ (0.00)	0.001*** (0.00)	0.001*** (0.00)
SCOP			−0.042* (0.02)
SCOP × capital	0.008 (0.01)	0.005 (0.00)	0.006+ (0.00)
Firm fixed effects	YES	YES	YES
N	30,470	7,293,067	7,295,242
Including number of SCOP obs	16,059	17,200	18,262

Note: Firm-clustered or individual-clustered standard errors in parentheses.
Firm-level observations include controls: firm age, workforce composition variables (proportion of managers and professionals, intermediate occupation, semi-skilled white-collar, unskilled white-collar, semi-skilled blue-collar, unskilled blue-collar, average employee age and age square and proportion of women), regional unemployment rate and year dummies.
Abbreviation: SCOP, *sociétés coopératives et participatives*.
+$p < 0.10$.
*$p < 0.05$.
**$p < 0.01$.
***$p < 0.001$.

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TABLE 6 Pay equation at the employee level, employee fixed effects.

	(1) **Matched sample** **Hourly pay (ln)**	(2) **Whole unmatched sample** **Hourly pay (ln)**
Demand shock	0.044** (0.01)	0.058*** (0.00)
SCOP × demand shock	0.043* (0.02)	0.017* (0.01)
Capital	0.004 (0.01)	0.003 (0.00)
SCOP × capital	−0013 (0.01)	−0.002 (0.009)
Firm fixed effects	NO	NO
Individual fixed effects	YES	YES
N	12,811	2,957,903
Including number of SCOP obs	7374	7412

Note: Individual-clustered standard errors in parentheses.

Controls: occupation dummies (six categories: professionals, intermediate occupation, semi-skilled white-collar, unskilled white-collar, semi-skilled blue-collar and unskilled blue-collar), age and age squared, tenure, firm size and year dummies.

Abbreviation: SCOP, *sociétés coopératives et participatives*.

$+p < 0.10$.

$^{*}p < 0.05$.

$^{**}p < 0.01$.

$^{***}p < 0.001$.

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TABLE 7 Hour equation at the firm level, firm fixed effect.

	(1) Matched sample	(2) Whole unmatched sample non-switchers	(3) Whole unmatched sample switchers
Demand shock	0.259*** (0.02)	0.278*** (0.00)	0.278*** (0.00)
SCOP × demand shock	0.075** (0.02)	0.044** (0.01)	0.030* (0.01)
Pay	−0.363*** (0.04)	−0.364*** (0.00)	−0.364*** (0.00)
Capital	0.014 (0.01)	0.023*** (0.00)	0.023*** (0.00)
SCOP			0.164+ (0.09)
SCOP × capital	−0.052*** (0.01)	−0.041*** (0.01)	−0.029*** (0.01)
SCOP × pay	−0.022 (0.05)	0.011 (0.03)	0.000 (0.03)
Firm fixed effects	YES	YES	YES
N	29,617	7,100,259	7,102,308
Including number of SCOP obs	15,567	16,660	17,685

Note: Firm-clustered standard errors in parentheses. Controls: firm age, workforce composition variables (proportion of managers and professionals, intermediate occupation, semi-skilled white-collar, unskilled white-collar, semi-skilled blue-collar, unskilled blue-collar, average employee age and age square and proportion of women), regional unemployment rate and year dummies.
Abbreviation: SCOP, *sociétés coopératives et participatives*.
+$p < 0.10$.
*$p < 0.05$.
**$p < 0.01$.
***$p < 0.001$.

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TABLE 8 Hour equation at the employee level, employee fixed effects.

	(1) Matched sample	(2) Whole unmatched sample	(3) Matched sample restricted to full-time workers
Demand shock	0.074***	0.065***	0.021**
	(0.02)	(0.00)	(0.01)
SCOP × demand shock	0.016	0.022*	0.020*
	(0.02)	(0.00)	(0.01)
Pay	−0.414***	−0.426***	−0.193***
	(0.06)	(0.00)	(0.03)
Capital	0.004	−0.000	0.003
	(0.07)	(0.00)	(0.00)
SCOP × capital	0.001	−0.008	−0.008
	(0.01)	(0.01)	(0.01)
SCOP × pay	−0.016	0.034	−0.006
		(0.04)	(0.03)
Firm fixed effects	NO	NO	NO
Individual fixed effects	YES	YES	YES
N	12,811	2,957,903	10,926
Including number of SCOP obs	7374	7412	6500

Note: Individual-clustered standard errors in parentheses.
Controls: occupation dummies (six categories: professionals, intermediate occupation, semi-skilled white-collar, unskilled white-collar, semi-skilled blue-collar and unskilled blue-collar), age and age squared, seniority, firm size and year dummies.
Abbreviation: SCOP, *sociétés coopératives et participatives*.
$+p < 0.10$.
$^{*}p < 0.05$.
$^{**}p < 0.01$.
$^{***}p < 0.001$.

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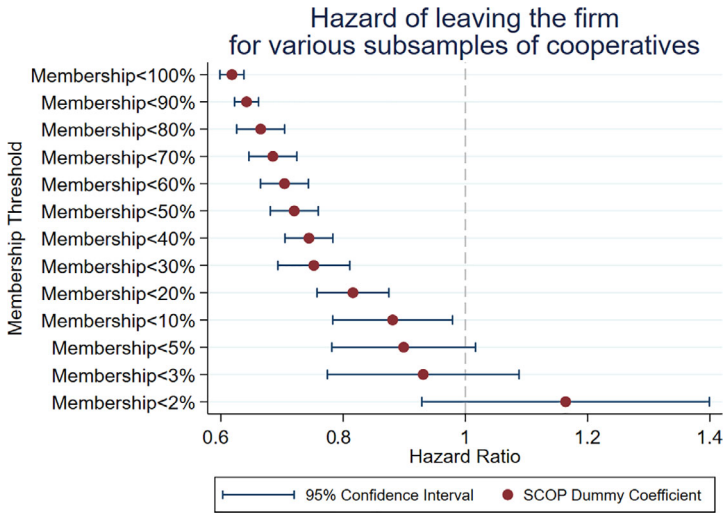


FIGURE 9 Hazard of leaving the firm for various subsamples of cooperatives. [Colour figure can be viewed at wileyonlinelibrary.com]

higher than 10%. Overall, SCOP employees' probability of leaving the firm is 25% (1–0.746) lower than CF employees'. If we only consider SCOPs where 20% of the workers are members or less, employees' probability to leave the firm is 18% (1–0.816) lower than in CFs and still significant at the 1% level. Only in SCOPs with 5% or lower membership does the probability of leaving the firm stop being significantly different from CF employees. This could indicate that a few SCOPs with very low membership rates behave like clubs and use non-member labour as their adjustment variable. However, only 20 SCOPs have a membership ratio lower than 5%.[27] More importantly, the results must be interpreted with caution because membership rates vary over time with changes in SCOP employment. Thus, SCOPs with very high membership may have recently dismissed non-members, for example.

4.4.1 | Women and men

In order to examine whether the greater employment stability in SCOPs affects both genders, we have estimated the firm-level employment equation separately for men and women. The results are presented in Table 9. Although the estimates are generally quite stable (except for that of pay in the matched-sample estimation), the SCOP effect for the demand shock is strikingly different for women and men, and this difference appears across samples. In general, demand shocks affect women's employment less than men's, both in SCOPs and CFs. However, the demand shock × SCOP interaction variable is only significant for men. Hypothesis H5 is confirmed: The impact of demand shocks on men's employment is reduced in SCOPs compared to CFs, but it is not the case for women. In all firms, when sales are 10% lower than the trend, men's employment decreases by 7.4%, whereas women's employment decreases by 6.5%. In SCOPs, women's employment also decreases by 6.5%, whereas men's employment decreases by 5.6%. The difference is strongly significant (z-tests for both coefficients are reported in Supporting Information Appendix A11).

[27] The details of numbers of observations and firms can be found in Supporting Information Appendix A10.

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TABLE 9 Employment equation at the firm level (separate estimations by gender, firm-level FEs, 2SLS and pay instrumented with lags 2 and 3).

	(1)	(2)	(3)	(4)	(5)	(6)
	Matched sample		Whole unmatched sample non-switchers		Whole unmatched sample switchers	
	Men	Women	Men	Women	Men	Women
	Employment (ln)	Employment (ln)	Employment (ln)	Employment (ln)	Employment (ln)	Employment (ln)
Demand shock	0.823***	0.754***	0.743***	0.654***	0.743***	0.654***
	(0.03)	(0.03)	(0.00)	(0.00)	(0.00)	(0.00)
SCOP × demand shock	−0.202***	−0.088*	−0.188***	−0.047	−0.150***	−0.053
	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Pay	−0.254	0.333	−0.239***	−0.175***	−0.239***	−0.175***
	(0.20)	(0.24)	(0.01)	(0.01)	(0.01)	(0.01)
Capital	0.130***	0.127***	0.171***	0.165***	0.171***	0.165***
	(0.02)	(0.02)	(0.00)	(0.00)	(0.00)	(0.00)
SCOP					0.890	0.673
					(0.72)	(0.68)
SCOP × pay	0.269	−0.769*	0.035	−0.183	−0.205	−0.261
	(0.34)	(0.37)	(0.27)	(0.27)	(0.26)	(0.26)
SCOP × capital	−0.019	0.042	−0.025	0.041	−0.018	0.025
	(0.03)	(0.03)	(0.02)	(0.03)	(0.02)	(0.02)
N	14,932	12,531	3,009,787	2,712,815	3,011,011	2,713,890
Including number of SCOP obs	8330	7015	8933	7581	9640	8205
Sargan overidentification test	3.533	3.232	7.388	2.593	7.118	2.777
	$p = 0.1709$	$p = 0.1987$	$p = 0.0249$	$p = 0.2736$	$p = 0.0285$	$p = 0.2495$

Note: Firm-clustered standard errors in parentheses. All models include the following controls: firm age, workforce composition variables (proportion of managers and professionals, intermediate occupation, semi-skilled white-collar, unskilled white-collar, semi-skilled blue-collar, unskilled blue-collar, average employee age and age square and proportion of women), regional unemployment rate and year dummies.

Abbreviation: SCOP, *sociétés coopératives et participatives*.

+$p < 0.10$.

*$p < 0.05$.

**$p < 0.01$.

***$p < 0.001$.

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TABLE 10 Hazard of leaving the firm for employees (Cox proportional hazard function [hazard ratios] with gender effects).

	Hazard ratio for leaving the firm
Demand shock	0.802***
	(0.02)
SCOP	0.803***
	(0.02)
Pay	0.859***
	(0.03)
High-skill job worker	0.966
	(0.03)
Men	0.975
	(0.02)
SCOP × men	0.892**
	(0.03)
N	144,223
Including number of SCOP obs	29,038

Note: Firm-clustered standard errors in parentheses. Included controls: occupation dummies (six categories: professionals, intermediate occupation, semi-skilled white-collar, unskilled white-collar, semi-skilled blue-collar and unskilled blue-collar), age and age squared, gender, seniority and firm size.
*** indicates 0.1% level of significance, **1%, *5% and +10%. All coefficients are hazard ratios. Significance tests are based on null hypothesis that hazard ratio = 1.0. All regressions include controls for worker occupation, firm capitalization, size of the firm and number of hours worked, as well as age and tenure of the workers.
Abbreviation: SCOP, *sociétés coopératives et participatives*.
$+p < 0.10$.
$^{*}p < 0.05$.
$^{**}p < 0.01$.
$^{***}p < 0.001$.

The employee-level estimations explaining the hazard of leaving the firm confirm this result (Table 10). In all firms, men have a lower probability to leave. This privilege is even greater in SCOPs. This confirms hypothesis H5 (men benefit from higher job stability than women, and this advantage is larger in SCOPs).[28]

4.4.2 | The role of skill

As with the effects of gender, we examine whether high- or low-skill jobs are generally more stable than average. Table 11 shows the results of separate estimations for the subsamples of high-skill and low-skill occupations. These estimates consistently show that demand shocks have a larger impact on low-skill workers' labour. However, we find employment stability is significantly greater in SCOPs only for workers in low-skill jobs, confirming hypothesis H6. In all firms, when sales are 10% lower than the trend, high-skill employment decreases by 6.1%, whereas low-skill employment decreases by 7.7%. In SCOPs, high-skill employment also decreases by 6.1%, whereas

[28] Another interesting hypothesis to test would be that of lower discrimination among firms with female top management or a high proportion of female members elected on the board (Dean, 2023). Unfortunately, this information is not available in the data.

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TABLE 11 Employment equation at the firm level (separate estimations for high- and low-skill occupations. 2SLS with firm fixed effects, pay instrumented with lags 2 and 3).

	(1)	(2)	(3)	(4)	(5)	(6)
	Matched sample		Whole unmatched sample non-switchers		Whole unmatched sample switchers	
	High-skill occupations	Low-skill occupations	High-skill occupations	Low-skill occupations	High-skill occupations	Low-skill occupations
	Employment (ln)	Employment (ln)	Employment (ln)	Employment (ln)	Employment (ln)	Employment (ln)
Demand shock	0.628***	0.907***	0.614***	0.777***	0.614***	0.777***
	(0.05)	(0.03)	(0.00)	(0.00)	(0.00)	(0.00)
SCOP × demand shock	0.093	−0.245***	−0.059	−0.215***	−0.031	−0.175***
	(0.07)	(0.04)	(0.04)	(0.03)	(0.04)	(0.03)
Pay	0.132	−0.394	−0.278***	−0.190***	−0.278***	−0.190***
	(0.45)	(0.31)	(0.03)	(0.02)	(0.03)	(0.02)
Capital	0.147***	0.156***	0.193***	0.176***	0.193***	0.176***
	(0.03)	(0.02)	(0.00)	(0.00)	(0.00)	(0.00)
SCOP					2.480*	0.560
					(1.01)	(0.67)
SCOP × pay	−0.842	−0.161	−0.977**	−0.107	−0.840*	−0.180
	(0.60)	(0.42)	(0.36)	(0.29)	(0.33)	(0.26)
SCOP × capital	0.039	−0.027	0.080*	0.001	0.060*	0.008
	(0.05)	(0.03)	(0.03)	(0.02)	(0.03)	(0.02)
N	6490	14,940	965,401	3,387,574	965,978	3,388,765
Including number of SCOP obs	4027	8153	4440	8754	4776	9439
Sargan overidentification test	1.547	6.319	10.83	7.482	11.10	7.252
	$p = 0.4614$	$p = 0.0424$	$p = 0.0582$	$p = 0.0237$	$p = 0.0574$	$p = 0.0266$

Note: Firm-clustered standard errors in parentheses. All models include the following controls: firm age, workforce composition variables (proportion of managers and professionals, intermediate occupation, semi-skilled white-collar, unskilled white-collar, semi-skilled blue-collar, unskilled blue-collar, average employee age and age square and proportion of women), regional unemployment rate and year dummies.

Abbreviation: SCOP, *sociétés coopératives et participatives*.

+$p < 0.10$.

*$p < 0.05$.

**$p < 0.01$.

***$p < 0.001$.

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low-skill employment decreases by 5.6%, and the differences are strongly significant (*z*-tests for both coefficients are reported in Supporting Information Appendix A11). This result is in line with Estrin (1991) who found Italian cooperative decisions to benefit the interests of blue-collar workers.

In estimations not reported here, we also ran separate regressions in order to measure any difference in wage and hours adjustments between different categories of workers. We find that hours are much more flexible for highly skilled workers in SCOPs than in CFs, but not significantly different for workers in low-skill jobs. Wages are more variable in SCOPs for all categories of workers, but the gap is larger for workers in high-skill occupations. Overall, SCOPs seem to use mostly highly skilled workers' wages and hours for adjustments, whereas CFs adjust mostly low-skill labour. This is partially in line with Lal (2023) who found that supervisors pay was more elastic to demand shocks in Indian worker cooperatives.

The results of the firm-level employment equation are confirmed by our estimates explaining the hazard of leaving the firm, with the hazard of leaving SCOPs lower than in other firms, but higher for high-skill employees, even though the hazard of leaving those jobs is generally lower[29] (Table 12).

5 | CONCLUSION

Although a large body of empirical work estimates the extent and value of firm-provided wage insurance (Guiso & Pistaferri, 2020), empirical assessments of adjustments along other firms' margins (employment) are still relatively scarce in comparison. Comparing democratic firms with conventional ones in a competitive environment, we can assess how much room for manoeuvre firms have in order to implement atypical human resource practices.

With a unique dataset covering all private-sector French firms with employees, we were able to compare adjustments made in response to demand shocks by worker cooperatives and CFs. In the context of mass unemployment in France and following several laws passed in 2016, 2017 and 2018 to facilitate redundancies, the question of the effect of firm governance on labour force adjustment seems particularly relevant. In addition, in the period covered by our data, the regulation of the French labour market provided considerable employment protection as well as a national minimum wage and binding industry-level collective bargaining with exceptionally broad coverage. Our study contributes to the recent empirical literature on the variety of firms' adjustment strategies, and the impact of governance and ownership structures on these strategies as well as, indirectly, to the study of the impact of employment and pay protection policies. We based our empirical approach on the extensive theoretical literature on LMFs and on recent empirical studies for Italy and Uruguay.

Our analyses make several important contributions. The first main objective was to verify international findings on the greater stability of employment and variability of pay in worker cooperatives, compared with other firms, for the French case. Our results show that despite the specific French institutional conditions that might dampen the effects of firm governance, employment stability in SCOPs is significantly greater than in CFs, whereas wages are significantly more flexible, in line with international evidence. When sales are 10% lower than the trend, firms lower

[29] We also carried out estimations looking at newly hired versus incumbent employees. These estimations yield inconclusive results, with new workers more likely to leave the firm with a negative demand shock, but with no difference between SCOPs and other firms, and inconsistent results across estimations regarding pay flexibility. Pay includes profit sharing distributed to all employees but often proportionally to tenure, which may explain the inconclusive findings.

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TABLE 12 Cox proportional hazard of leaving the firm, employee level, with skills effects.

	(1) Hazard ratio for leaving the firm
Demand shock	0.812*** (0.02)
SCOP	0.678*** (0.01)
Pay	0.766*** (0.02)
High-skill job worker	0.854*** (0.02)
SCOP × high-skill	1.307*** (0.05)
Men	0.931*** (0.01)
SCOP × men	0.812*** (0.02)
N	144,223
Including number of SCOP obs	29,038

Note: Firm-clustered standard errors in parentheses. Included controls: occupation dummies (six categories: professionals, intermediate occupation, semi-skilled white-collar, unskilled white-collar, semi-skilled blue-collar, unskilled blue-collar), age and age squared, gender, seniority and firm size. *** indicates 0.1% level of significance, **1%, *5% and +10%. All coefficients are hazard ratios. Significance tests are based on null hypothesis that hazard ratio = 1.0. All regressions include controls for worker occupation, firm capitalization, size of the firm and number of hours worked, as well as age and seniority of the workers.
Abbreviation: SCOP, *sociétés coopératives et participatives*.
$+p < 0.10$.
$^{*}p < 0.05$.
$^{**}p < 0.01$.
$^{***}p < 0.001$.

employment by 7.8% on average, but only by 6.2% in SCOPs. With the same 10% decrease in sales, pay is 0.36% lower in firms on average but 0.47% lower in SCOPs. In addition to estimating firm-level equations, we use a Cox proportional hazard model to measure the individual hazard for employees of leaving their firm, controlling for idiosyncratic firm demand shocks as well as detailed individual characteristics, and estimate pay equations at the employee level. The employee-level results confirm the findings from our estimations at the firm level for both SCOPs' greater employment stability and pay variability, making our results very robust. However, we find only weak evidence that hours may be more of an adjustment variable in worker cooperatives than in other firms. One avenue we have not been able to explore is the influence of technological specification on the adjustment strategies of SCOPs. It is possible that lower employment responsiveness in SCOPs reflects the use of labour-enhancing technology. This would be worth exploring in future research with precise data on technological specifications.

Our second key objective was to examine whether workplace democracy resulted in a more equitable distribution of adjustments in SCOPs than in other French firms. Going beyond formal governance, which gives equal voting rights to all members in cooperatives, we considered the possibility that adjustments differed across categories of workers in SCOPs due to majority decision rules and common prejudices. We find men's employment is more stable in SCOPs than in

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CFs, but women's employment flexibility is more similar in SCOPs and CFs. This finding suggests that international evidence on greater employment stability in worker cooperatives may be incomplete, and further analyses should look at the effects of gender, as well as other discriminated characteristics such as ethnic background and disability, on employment adjustments. Finally, in line with existing studies proposing worker cooperatives pursue the interests of median members in low-skill jobs, we find that low-skill employment is more stable in SCOPs than in CFs, but high-skill employment is just as flexible in SCOPs as in CFs. When it comes to employment adjustments, SCOPs do not seem to behave systematically as clubs for a few privileged members, but the job stability they promote is not universal.

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Additional supporting information can be found online in the Supporting Information section at the end of this article.

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